

IT Service Desk Model Literature Review: Benefits and Challenges

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Abstract—IT service desk is an important element in IT service management. As a single point of contact (SPOC), IT service desk receives incident and fulfill the requests. Many governments are still having difficulties in various areas, particularly during implementation. The management of incident reports and service requests in the government is only submitted directly to everyone, giving rise to multiple points of contact. Meanwhile, a standard or framework cannot be directly adopted but requires adjustments to the characteristics of the organization. Through this research, the challenges, and benefits of implementing IT service desk were identified. Literature review using SLR was carried out. A total of 40 articles from four well-known databases were selected for review. The results show that some of the benefits obtained from implementing the IT Service desk include significant improvement in process metrics, reduction in IT costing, better documentation, increased employee and customer satisfaction, reduction in staff, improvement of IT service quality, and internal communication efficiency. Meanwhile, the challenges include large investment required, organizational resistance, difficult planning to implement the existing framework, complexity of the framework, lack of internal management support, difficulty in measuring benefits, lack of knowledge/skills, process evaluation (expensive and time consuming), limited of resources, and gradual cost reduction.

Keywords—benefits; challenges; IT service desk; SLR

I. INTRODUCTION

For most companies and major corporations, Information technology (IT) has become a very important part of every business plan. From the 1980s to the present, IT has been used as an important support for the organization's business operations [1]. So that the organization's dependence on IT is getting bigger and effective management is needed to manage it [2]. Therefore, IT management gets a lot of attention because it relates to IT costs [3].

Organizations need to carry out effective and efficient IT management to improve the IT services provided to users. In this case, to manage IT services, organization might use Information Technology Service Management (ITSM). The IT service desk focuses on IT operations for the organization, more specifically service delivery and organizational support [4]. ITSM takes a strategic approach, design, service delivery, and improvement for organizational needs [5]. ITSM does not only focus on technology but focuses on customer service [6].

IT service desk is one of the important elements in ITSM, especially as a single point of contact (SPOC) for receiving

incidents, recording incidents, providing solutions to incident problems, and escalating incidents to second-line support or third-line support if the service desk cannot handle incidents [7]. The implementation of the service desk is expected to improve service performance and ensure IT services run well.

Currently, various standards and frameworks on ITSM are available. Starting from ISO/IEC 20000, ITIL, and COBIT to service desk standards. [5] states that, from the results of a survey conducted in 2017, the most widely used ITSM framework or best practice is ITIL [8]. Although there is quite a lot of information related to the ITSM framework and standards for organizations or companies, several companies, both governments, private and state-owned have not been certified in IT Service Management [9].

However, government, as it is known, has unique characteristics, ranging from organizational hierarchy, government regulations, and availability of human resources, to organizational culture. The Indonesian government itself has a legal umbrella to regulate service management, one of which is Presidential Regulation Number 95 of 2018 concerning Electronic-Based Government Systems [10]. Meanwhile, a standard or framework cannot be directly adopted but requires adjustments to the characteristics of the organization. Some argue that the existing framework is difficult to execute [11], citing the suggested model's significant complexity and in terms of content knowledge, does not yield the expected benefits. Furthermore, the successful implementation of a standard in one company does not necessarily follow when it is implemented in another. As a result, benefits, problems, and practices for IT Service Desk installation were extracted, critically reviewed, and explored in this study.

II. METHOD

This section describes the Systematic Literature Review (SLR) approach. An overview of the IT Service Desk section is offered in all the selected articles. SLR summary is conducted in the concluding section.

A. Research question

Research questions that will be answered in order to obtain research objectives include:

RQ1: What are the main IT Service Desk benefits?

RQ2: What are the challenges faced in implementing IT Service Desk?

B. Search process

The search process begins by specifying the repository database. There are 4 repository databases which are presented in TABLE 1. Continue using SLR and gather accurate data to achieve the set purpose of addressing the formulated research questions. The search was carried out for 2 weeks from March 31, 2022 to April 14, 2022.

TABLE 1 DATABASE REPOSITORIES

Database	URL
Science Direct	https://www.sciencedirect.com/
Emerald Insight	https://www.emerald.com/insight/
IEEE	https://ieeexplore.ieee.org/Xplore/home.jsp
Google Scholar	https://scholar.google.com/
Wiley Online Library	https://onlinelibrary.wiley.com/
Springer	https://link.springer.com/

From several journal repositories, the selection of specific journals using special keywords is used together with AND and OR operators [12]. These keywords include:

“IT Service Desk” AND

(“Advantages” OR “Benefits” OR “Challenges” OR “Difficulties” OR “Standards” OR “Framework” OR “Disadvantages” OR “Barriers” OR “Models”).

Only English-language articles and publications published in journals or conference proceedings are considered for review. Finally, no date filter is applied. The first stage is to search for articles using important keywords without any filters applied in each repository. After that, three filters were developed. Since each electronic repository has its own "search technique", each repository is assigned its own keyword adaptation.

As previously stated, the selected articles will be filtered three times in each repository. TABLE 2 shows the filter process for each selected repository along with the number of results.

TABLE 2 RESULT OF THE FILTERING ARTICLE

	No Filter	1 st Filter	2 nd Filter	3 rd Filter
Science Direct	6903	189	143	11
Emerald Insight	200	45	28	5
IEEE	1104	20	5	4
Google Scholar	1337	35	20	10
Wiley Online Library	300	50	10	8
Springer	305	50	20	12
Total	10149	389	226	50

Additionally, items from the Google Scholar repository are added manually. Since this repository cannot extract the entire article collection at once, the first filter is not used.

The first filter was created to select articles that only mention the IT Service Desk from the principles contained in the main article. This method is done by only looking at the title, abstract, and keywords written in the abstract. These three things are seen because they are the most important part in summarizing the topic of the article. It is worth rejecting a

large number of articles with this filter. Then a second filter is performed to remove duplicate items. thus, 226 papers were viewed and selected using the inclusion criteria mentioned above in the third filter, resulting in a final 40 articles.

III. RESEARCH MAPPING

A total of 40 items were included in the final package. Only articles from recognized publications were accepted, as previously stated. As seen in Fig. 1, which shows the keyword mapping of the articles utilizing the VOS Viewer and Publish or Perish tools.

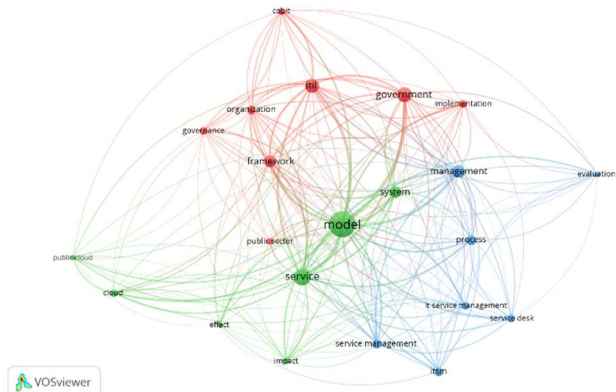


Fig. 1 Research Mapping based on keyword

Fig. 2 depicts the year-by-year distribution of articles. The year interval runs from 2006 to 2022, as may be seen. Considering the evolution of articles through time and the study's article sample, it can be stated that the most research related to ITSM in 2018. While in other years it was volatile. It is implying that ITSM concept research is interesting.

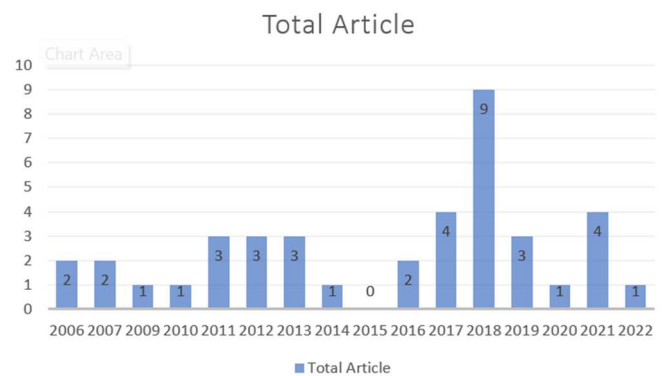


Fig. 2 Articles by Years

IV. RESULT AND DISCUSSION

A. IT Service Desk Benefits

The IT service desk provides several benefits to the organization if it is implemented in a planned and structured manner. As previously stated, the SLR methodology was used. TABLE 3 summarizes and characterizes the benefits derived from the selected research studies.

Greatest IT firms employ ITSM frameworks because they produce the most visible improvements in the organization [2]. A significant number of these recognizable increases happen at the functional level [24], where process metric upgrades reported by the association are promptly clear.

TABLE 3 IT SERVICE DESK BENEFITS

No	Benefits	Articles
1	Significant improvements in process metrics (i.e., incident handling time, controlled errors, predictable problems)	[2] [13] [14] [15]
2	Reduction in IT costing	[2]
3	Better documentation	[16]
4	Employee satisfaction increases	[16] [17]
5	Reduction in staff	[16] [18]
6	Higher customer satisfaction	[19] [17] [20] [18] [15]
7	IT service quality improvement	[21]
8	Higher efficiency/performance	[22] [23] [18] [15]
9	Internal communication process becomes efficient	[22]

Besides, it is contended that this sort of advantage improved framework consistency, bringing down server or application issues [25]. Moreover, some review demonstrates that the volume of occurrences and mistakes is decisively diminished on the grounds that [19] with processes set up, issues that might create during routine tasks can be managed proactively.

Furthermore, being proactive in managing bothersome and troublesome occasions that might emerge may lessen monetary waste [2], making this advantage one of the most huge for upper administration. According to the articles, utilizing ITSM standards can bring about a huge decrease in activity and support costs, as well as in help arrangement costs [21], which are inseparably connected with the improvement of framework and functional proficiency [16].

IT Service Desk helps companies in handling effective service management. Indeed, it is often stated in articles that this framework assists Management in producing better process documentation [16] and monitoring, enabling more complete IT process audits and reports [19]. Furthermore, better documentation and monitoring results in increased transparency and comparability for the organization. Therefore, in this case process maturity can be increased [21] and allow internal companies to manage their processes, thus supporting organizations operating in an uncertain environment [24].

A few researcher likewise guarantee that ITSM structures work on the straightforwardness and heartiness of inside organization tasks [16]. Employee satisfaction will rise as a result of improved and deeply grounded "conventions", which will help IT employees meet expectations [16]. Employee satisfaction and confidence not only help firms in reorienting or changing their way of life, yet in addition improves organization efficiency.

Probably the most combative advantage talked about in the examination is staff decrease. Staff can be moved or even reduced, according to one report, because through automation of some or all of the framework capabilities can increase process maturity [16]. Indeed, various attempts have been made to automate ITSM operations [26], in view of the worry of moving individuals. [27].

The scientific community has emphasized that better process control and performance can improve service quality, in which case this is a general benefit. As a result, quality improvement leads to the benefit most needed by the company: customer satisfaction [11]. Consumer satisfaction is defined as an increase in satisfaction of current and future customer needs as the quality of framework operations improves [21]. Furthermore, these frameworks provide consumers with increased availability, reactivity, and expertise [16].

Furthermore, according to some experts, a continuous-improvement strategy improves the quality of IT services offered [21], increasing the adaptability and flexibility of IT services [28].

In reality, firms looking to increase the effectiveness of their functions are looking into ITSM methodologies [22]. Businesses' processes are said to mature as a result of ITSM, making operational activity performance more stable [1]. Adapting organizational practices to newly implemented procedures can also help businesses boost productivity.

Last but not least, by Communication is improved across all organizational tiers when ITSM frameworks and procedures are well-established and defined, resulting in a "single language" [22]. This method promotes internal organization consultation [16] and boosts the efficacy and service quality of IT suppliers.

B. IT Service Desk Challenges

Although the advantages of having an IT Service Desk, many organizations experience issues on a regular basis. TABLE 4 lists and analyses the challenges discovered in the literature.

TABLE 4 IT SERVICE DESK CHALLENGES

No	Challenges	Articles
1	Requires a big investment	[3]
2	Organization resistance	[29] [30] [13] [31]
3	Hard of planning to implement multiple frameworks	[29]
4	Frameworks complexity	[32] [30] [20]
5	Lack of management support	[32] [30]
6	Difficult on quantifying the benefits	[32]
7	Lack of knowledge/skills	[33] [17] [30] [34] [23]
8	Processes evaluation (expensive and time-consuming)	[35]
9	Limited resources	[36] [18]
10	Steady lower costs	[37]

IT service costs are expected to account for 70% to 80% of IT organization costs [38], requiring a considerable investment and providing a significant organizational challenge. The framework needs to be introduced, so there will also be a necessary expenditure as there will be a change from the organizational "style of function" [3]. Not only that, but there must also be costs and investments taken into account for maintaining the necessary infrastructure. And costs related to performance monitoring and evaluation of continual improvement are also considered [11].

The key issues in ITSM can be separated into four areas, according to Keel et al. [2]: technology, data, process, and people. The most reported concern is coworker opposition to change. This reluctance is linked to the difficulty of adopting and putting in place a new strategy at the corporate level, according to corporate culture [29]. Humans or peoples there will be rejection, thus causing psychological problems that have an impact on management. Thereby causing delays in implementation. In summary, before making major changes, management should prepare a technical team first. Such problems develop because of organizational culture lacking secondary values such as good innovation and support for change.

Several organizations are required to use a variety of frameworks/standards in order to comply with legislation and improve their processes [32]. In fact, enterprises are finding it difficult to design these deployments, because they must examine the two frameworks' integration and interoperability for cost savings, reduced complexity, and efficiency [29]. Furthermore, some frameworks' processes overlap necessitating the use of a suitable strategy while using numerous frameworks.

ITSM frameworks have often been chastised for being overly complicated. The broad and generic definitions of the frameworks are cited by some of the critics. Furthermore, there is no implementation guidance in the frameworks and standards [36]. Furthermore, ITSM does not provide a way to measure and improve processes, but only defines the flow and interactions between processes [32].

TABLE 4 shows that the majority of the issues relate to the implementation phase of the framework. Despite their growing popularity, the challenge for IT companies or organizations that use IT is that they still lack clear implementation guidelines in the application of the ITSM framework [39].

A lack of management assistance was also mentioned in the literature as a problem. a case study said, although enthusiastic support from the "management layer" for implementation was high, the commitment was insufficient [12]. Ensured that the implementation failed. Not only that, but limited management support can also cause delayed implementation not according to plan [32], resulting in higher expenses. Furthermore, without senior management's backing, the execution could fall short of the organization's goal and strategy.

It may be difficult to justify the cost of adoption and investment in a framework by attempting to illustrate the benefits. Despite the potential benefits, literature suggests that framework implementation might be hampered, not only because of the financial outlay, but also because of the difficulties in achieving the benefits and establish a link between operational and revenue "performance measures". Quantifying the benefits can be difficult, and this might lead to a lack of ongoing commitment [32].

Furthermore, enterprises face a major problem in terms of ITSM framework knowledge and skills [33]. "The gap between knowledge of the ITSM framework and its application" is mentioned in the literature, it appears that the challenges faced are closely related to complex frameworks. In addition, the lack of experience between project managers and other key organizational members makes the implementation and maintenance of procedures problematic.

Process evaluations are a difficult issue for many IT firms that have deployed an ITSM framework, but they are critical for continual ITSM improvement [40]. Process evaluation methods, on the other hand, are costly and time-consuming [35]. Furthermore, the frameworks lack a process improvement assessment method that may aid organizations [32]. According to some researchers, an organization without an evaluation process will fail and be unprofitable in the long run, because no improvement strategy will be implemented, and the current state of activity is unclear.

According to this study, the lack of resources difficulty is due to a lack of management support, the problem has become even worse. A fundamental challenge on framework implementation is calculate the "quantity" of resources to be supplied [36]. Gathering enough resources for process rollout and maintenance can be difficult without managerial support.

Finally, not only for income, but also for continuous ITSM improvement, increasing and maintaining the quality of delivered services is a critical duty. It has been suggested, however, that maintaining constant or reduced costs while improving delivery quality is difficult [37].

V. CONCLUSION

Some of the benefits obtained from implementing the IT Service desk include significant improvement in process metrics, reduction in IT costing, better documentation, employee satisfaction increases, reduction in staff, higher customer satisfaction, IT service quality improvement, higher efficiency/performance, and the internal communication process becomes efficient.

Despite the many advantages of IT service desks, organizations nonetheless face a variety of issues due to a lack of clear norms in specific situations. The challenges faced from implementing it service desk include large investment required, organizational resistance, difficult planning to implement the existing framework, complexity of the framework, lack of internal management support, difficulty in measuring benefits, lack of knowledge/skills, process evaluation (expensive and time consuming), limited of resources, and gradual cost reduction.

Given the requirement to design improvement plans to develop mature processes that will deliver predictable benefits, the lack of guidelines and standards for ideal processes and process assessments may constitute the "follow-up work" in this section. Considering this, research into novel solutions for process evaluation and improvement should be addressed. In addition, new studies should entail a thorough evaluation of the link between benefits and challenges.

VI. REFERENCES

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